

FARBOD HAERI

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SUMMARY

Mechanical engineering student on the combined B.S./M.S. track, specializing in controls and robotics, with experience in UAV research, mechanical design and fabrication, and the controls and software that make hardware actually work. Interested in flight control, state estimation, and robotics for autonomous systems.

SKILLS

Programming: Python, MATLAB, C++, Java, JavaScript, HTML/CSS, Flask, Git

CAD / CAM / CAE: SolidWorks (Simulation, PDM), Fusion 360, NX, AutoCAD, ANSYS (FEA)

Manufacturing: CNC mill, manual lathe, waterjet, laser cutter, FDM 3D printing

Electronics: wiring, power distribution, motor / ESC integration, flight-controller (Pixhawk) bring-up, soldering (through-hole + SMD)

Controls: PID, lead-lag, notch filtering, state-space / LQR, loop shaping, pole placement, sensor fusion

Test & Metrology: optical metrology, machine vision, fatigue testing, defect quantification, dataset auditing

RESEARCH EXPERIENCE

Undergraduate Researcher, UAV Controls & Robotics

Jan 2026 – Present

Prof. Thomas Bewley · UC San Diego

San Diego, CA

- Designing and building a fixed-tilt hexacopter, fully actuated in six degrees of freedom: its rotors sit at fixed cant angles, so it slides in any direction and holds a steady attitude instead of tilting to move like a normal multirotor. Built for target tracking and tight-space search.
- Own the build end to end: designed the airframe in SolidWorks, printed it as a single 3D-printed monocoque to keep it light and stiff, and assembled two flying prototypes.
- Ran thrust-stand tests comparing ducted and un-ducted propellers, measuring thrust, torque, and efficiency, and used the data to settle the propulsion and ducting instead of guessing at it.
- Designing the 6-DoF flight controller that decouples translation from rotation; deriving and tuning the gains in MATLAB, then checking attitude and position control against motion-capture ground truth.

Undergraduate Researcher, Advanced Materials & Metrology

Aug 2025 – Dec 2025

Prof. Maziar Ghazinejad · UC San Diego

San Diego, CA

- Built MATLAB pipelines turning raw optical-metrology and machine-vision images into defect-fraction and surface-roughness data, feeding clean, repeatable numbers into fatigue-life models for additively-manufactured steels.
- Designed 10+ mechanical components and machined the fatigue and metrology fixtures myself on the CNC mill and waterjet.
- Ran fatigue tests across material variants, logged the stress data, and audited the datasets so every result stayed traceable.

PROJECTS

Automated Cognitive-Training System for Mice | *Senior Design, Jared Young Lab, UCSD*

Jan 2026 – Jun 2026

- Built the operator dashboard (HTML / CSS / JavaScript) the lab uses to launch sessions and watch live per-mouse reaction time, accuracy, and performance across multiple cages.
- Wrote the Python / Flask supervisory software, a training-cycle state machine and per-mouse scheduler that runs every stage from habituation through the 5-choice task, plus the data-logging and the Wi-Fi / MQTT link to the touchscreen rigs, so protocols change in code instead of re-flashing hardware.

MAE3 Autonomous Robot | *1st Place, UCSD MAE3 Robot Competition*

Sep 2024 – Dec 2024

- Led mechanical design on a small team; built the chassis and drivetrain from scratch and iterated through three revisions to take first place.

WORK EXPERIENCE

Student Engineer Intern

May 2024 – Aug 2024

County of Sacramento

Sacramento, CA

- Benchmarked energy use across 120+ county buildings, scoring each building's efficiency and rebuilding 50 buildings' utility histories into clean baselines to flag the low performers for audit and energy reduction.

Materials Manager

Jun 2022 – Apr 2024

Apple

Elk Grove, CA

- Planned and kitted inventory across 500+ SKUs for new-product builds, flashed and tested production units, and escalated supply risks to keep lines from stopping.

EDUCATION

University of California, San Diego

San Diego, CA

M.S. Mechanical Engineering

2026 – 2027

B.S. Mechanical Engineering, Specialization in Robotics & Controls

2024 – 2026